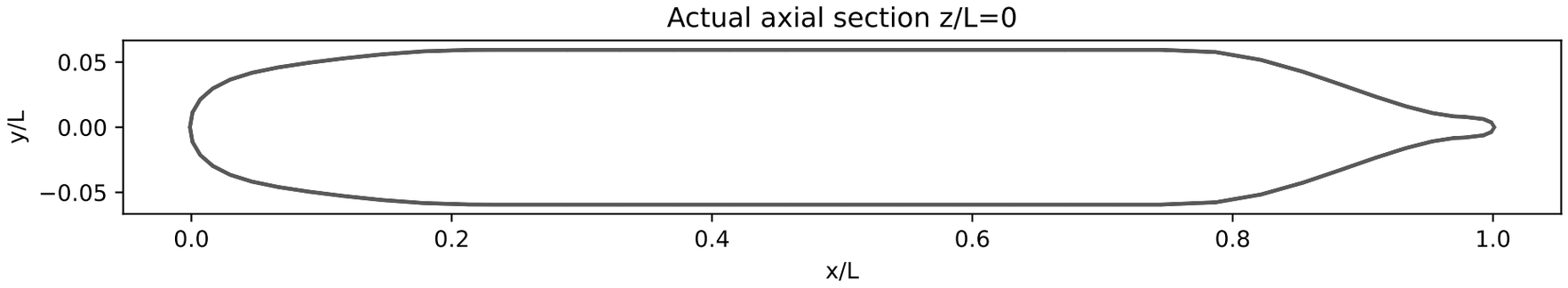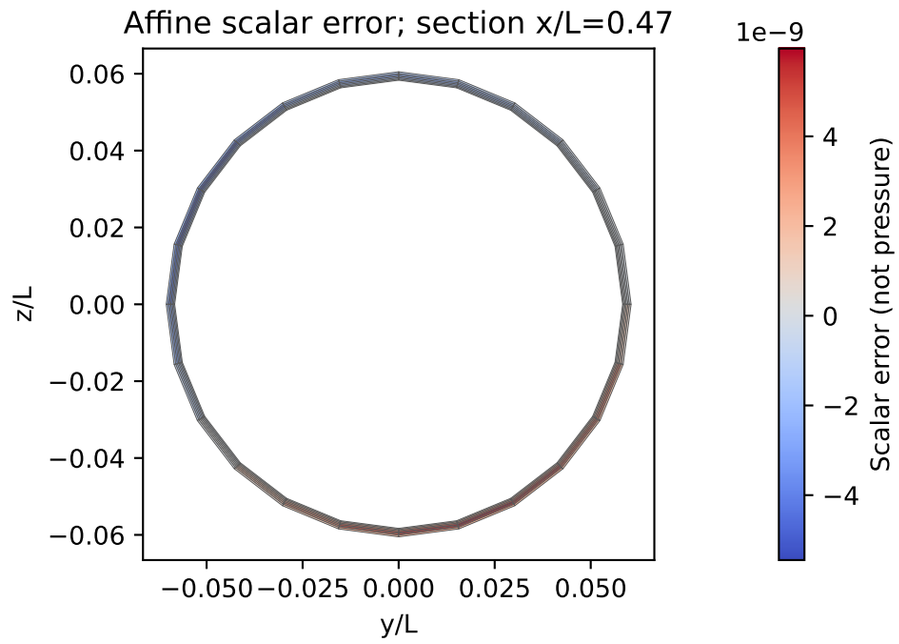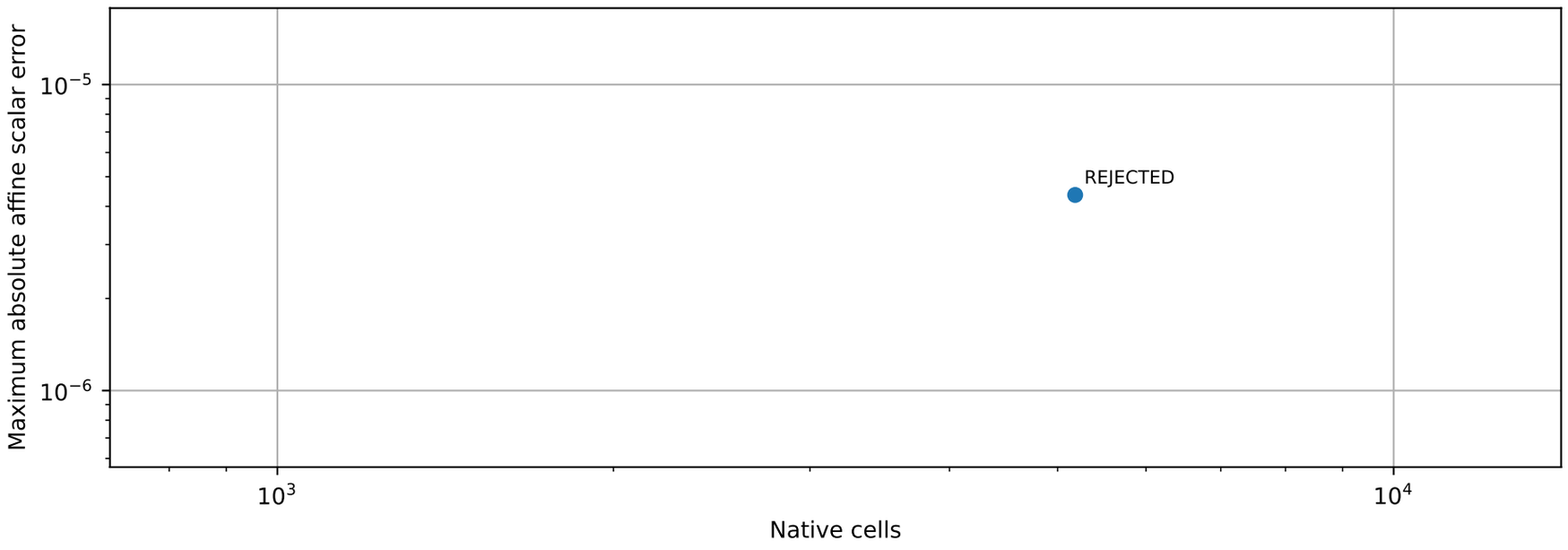SUBOFF thin-layer scalar control — no exterior flow or Cp/Cd